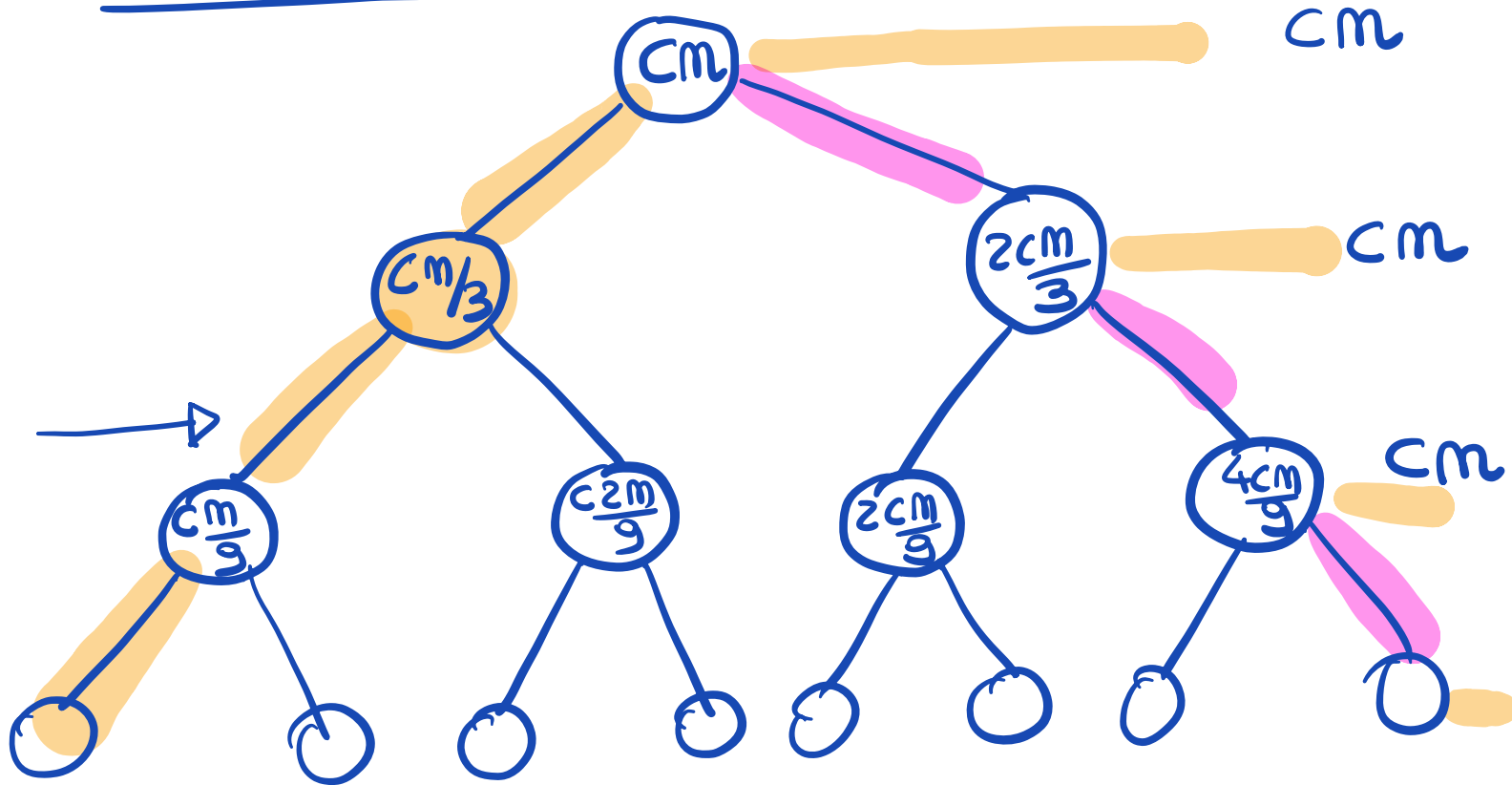


$$T(n) = \begin{cases} 1 & \text{se } n=1 \\ T(n/3) + T(2n/3) + cn & n > 1 \end{cases}$$

ALBERO DI RICORSIONE



$$\left(\frac{1}{3}\right)^i \cdot n = 1$$

$$\left(\frac{2}{3}\right)^i \cdot n = 1$$

$$\log_3 n$$

$$\log_{3/2} n$$

$$T(n) \leq \sum_{i=0}^{\log_{3/2} n} cn = cn \sum_{i=0}^{\log_{3/2} n} 1 = cn \log_{3/2} n$$

$$O(n \cdot \log n)$$

$$T(n) \geq \sum_{i=0}^{\log_3 n} cn = \left[cn \sum_{i=0}^{\log_3 n} 1 \right] = cn \cdot \log_3 n$$

$$\Omega(n \cdot \log n)$$

$$\Theta(n \cdot \log n)$$

$$T(n) = \begin{cases} 1 & n=1 \\ 3T(n/4) + \boxed{cn^2} & n > 1 \end{cases}$$

$\boxed{cn^2}$ confronto con $n^{\log_4 3} \approx \boxed{n^{0.8}}$

$$cn^2 = \Omega\left(n^{\log_4 3 + \varepsilon}\right) \quad \varepsilon = 0.2$$

CASO 3

$$\boxed{a f\left(\frac{n}{b}\right) \leq d f(n)}$$

$$3 \cdot \cancel{\frac{cn^2}{4^2}} \leq d \cdot \cancel{cn^2}$$

$$\Theta(\cancel{cn^2}) \quad \Theta(n^2) \cdot \frac{3}{4^2} \leq d$$

$$\boxed{d = \frac{3}{4^2}}$$

Max(A, m)

{ if m=1 then return A[1]

else

m = Max(A, m-1)

if A[m] > m

then return A[m]

else return m

}

$$T(m) = T(m-1) + C$$

$$T(m) = \underbrace{T(m-1)}_{\swarrow} + C = \underbrace{T(m-2)} + C + C$$

$$T(m-1) = \underbrace{T(m-2)} + C \quad \underbrace{T(m-3)} + C + C + C$$

$$T(m-2) = \underbrace{T(m-3)} + C$$

$$T(m-4) + C + C + C + C$$

$$T(m-3) = T(m-4) + C$$

$$T(1) + \underbrace{C + C + C + \dots + C}$$

$$\Theta(m)$$

$$T(1)$$

$$O(m)$$

$$T(1) = C$$

Algo(A, x, i, j)

{ if $i > j$ then return -1

else

$$m = (i + j) / 2$$

if $A[m] == x$

then return m

else if $A[m] > x$

then return **Algo**(A, x, i, m-1)

else return **Algo**(A, x, m+1, j)

}

$$T(m) = T(m/2) + C$$

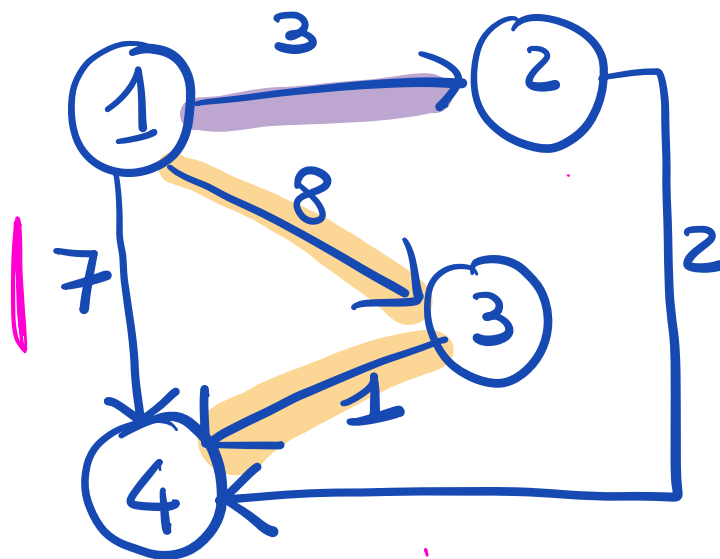
$$+ 1$$

$$+ \Theta(1)$$

$$\boxed{C} \quad m^{\log_a b} = m^{\log_2 1} = m^0 = \boxed{1}$$

CASO 2

$$\Theta(m^{\log_b a} \cdot \log m) = \Theta(\log m)$$



$d^{(4)}$

	1	2	3	4
1	0	3	8	5
2	∞	0	∞	2
3	∞	∞	0	1
4	∞	∞	∞	0

$\pi^{(4)}$

	1	2	3	4
1	NIL	1	1	2
2	NIL	NIL	NIL	2
3	NIL	NIL	NIL	3
4	NIL	NIL	NIL	NIL

$d^{(1)}$

$$\begin{array}{l} d_{11} + d_{11} \\ d_{11} + d_{21} \\ d_{11} + d_{31} \\ d_{11} + d_{41} \end{array}$$



$$\begin{array}{l} d_{12} + d_{11} \\ d_{12} + d_{21} \\ d_{12} + d_{31} \\ d_{12} + d_{41} \end{array}$$

passo 0

$$\begin{array}{l} d_{13} + d_{11} \\ d_{13} + d_{21} \\ d_{13} + d_{31} \\ d_{13} + d_{41} \end{array}$$

passo 1

